

Samuelson Rule

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Samuelson (1954) rule : $MRS_{gc} (g/c) = 1$
 $[\partial u / \partial g = \partial u / \partial c]$

Samuelson spending : $(g/c)^*$ st $MRS_{gc}((g/c)^*) = 1$
 Amount of public spending that satisfies Samuelson rule $\rightarrow$ optimal public spending in a neoclassical model

First-order approximation of MRS_{gc} around $(g/c)^*$:
 function g/c

$$MRS_{gc} \approx MRS_{gc}(g/c^*) + \frac{dMRS_{gc}}{d(g/c)} \times [g/c - g/c^*]$$

(omit all terms of order 2 & above) $[df = f'(x) \cdot dx]$

1

$$\frac{1}{\epsilon} = - \frac{d \ln MRS_{gc}}{d \ln g/c}$$

$$= - \frac{g/c}{MRS_{gc}} \times \frac{d MRS_{gc}}{d g/c}$$

derivative evaluated at g/c^* :

$$\frac{d MRS_{gc}}{d g/c} = - \frac{1}{\epsilon} \cdot \frac{MRS_{gc}(g/c^*)}{g/c^*} = - \frac{1}{\epsilon} \cdot \frac{1}{g/c^*}$$

$$\Rightarrow MRS_{gc} = 1 - \frac{1}{\epsilon} \cdot \frac{g/c - g/c^*}{g/c^*}$$

$$1 - MRS_{gc} = \frac{1}{\xi} \cdot \frac{g/c - g/c^*}{g/c^*}$$

elasticity of substitution
b/w
goods

departure from
Samuelson spending
= stimulus spending